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The dignity experience of people with disability when using trains and buses in an Australian city

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ABSTRACT

When transport systems are accessible and inclusive, people with disability experience dignity. Alternatively, when personal mobility is constrained by physical, social and/or communication, barriers, people with disability experience exclusion and an increasing vulnerability to indignity. This study sought to qualitatively explore the role of trains and buses in an Australian city in supporting access, inclusion and dignified mobility experiences for people with disability. Twenty-six semi-structured interviews were conducted with participants with diverse visible and invisible disabilities and were analysed thematically using Framework Analysis. The findings highlight the complexities involved with navigating public transport systems while maintaining dignity for people with disability. Accessible and inclusive information, infrastructure, and interactions with staff ensured dignified mobility experiences. If any part of a journey was inaccessible, participants were vulnerable to indignity. Dignified mobility experiences represent a complex and dynamic interaction between personal experiences and preferences, impairment-specific requirements, transport infrastructure, interpersonal experiences, and information inclusivity.

ARTICLE HISTORY

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KEYWORDS

Dignity; accessibility; inclusion; people with disability; public transport

Points of interest

- Public transport systems can play an important role in enabling, facilitating and sustaining dignity for people with disability.
- People with disability experience theoretical and practical gaps when using trains and buses.

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- Gaps result from infrastructure, environment, information, and attitudinal barriers.
- Universal Design, access to accessible and inclusive information, and empathic attitudes help create dignified mobility experiences for people with disability when using buses and trains.

Introduction

Public transport systems are important mechanisms for promoting freedom of movement, enabling people with disability to participate in society in self-determined ways and with dignity. Despite this, more than 30% of people with disability in Australia experience difficulties using public transport (Unsworth et al. 2019; Unsworth et al. 2021). People with physical, cognitive, and sensory impairments experience greater difficulty using public transport, often because of precincts and environments that are ill-equipped or inaccessible (Almada and Renner 2015; Calle et al. 2022; Kersten et al. 2020).

As a signatory to the *United Nations Convention on the Rights of Persons with Disabilities* (UNCRPD) (CRPD (Committee on the Rights of Persons with Disabilities) 2006), Australia must recognise that freedom of movement is a right for people with disability. The UNCRPD establishes the human rights imperative that people with impairment are included, with dignity, at all levels of society, specifically referring to the importance of accessibility, inclusion, and Universal Design in Articles 9 and 20 (CRPD (Committee on the Rights of Persons with Disabilities) 2006). Signatories of the UNCRPD must 'ensure personal mobility with the greatest possible independence for persons with disabilities' (CRPD (Committee on the Rights of Persons with Disabilities) 2006, Article 20) to enable 'full and effective participation and inclusion' for people with disability (CRPD (Committee on the Rights of Persons with Disabilities) 2006, Article 3).

A critical impediment to achieving accessibility and inclusion for Australians with disability is the pervasive nature of the medical model of disability. The medical model constructs disability as an individual problem, flaw, or deficit whereby the individual takes action to remedy or cure their impairment (Levine and Karner 2023). Considerations for accessibility and inclusion are often omitted from transport planning and design because disability stems from an individual deficit or impairment (Levine and Karner 2023). The everyday ableism and exclusionary processes embedded in Australia's public transport systems are typically invisible to non-disabled people, are somewhat culturally accepted, and increase the social exclusion of potential transport users with disability (Calder-Dawe, Witten, and Carroll 2020; Calle et al. 2022; Levine and Karner 2023; Tillmann et al. 2013). Contemporary understandings of disability, which establish disability as a mismatch between

impairment and environment (built, virtual, social, etc.) and centres on the inherent dignity of each person, guide this research to better understand the impact of lived experience on accessibility and inclusion (Lawson and Beckett 2021)

Public transportation impacts on freedom of movement in urban spaces and quality of life for all people, but especially for marginalised people (Calle et al. 2022). Individual mobility can be constrained by intersectional and overlapping layers of disadvantage (Meekosha and Shuttleworth 2009). These intersectional factors, like diversity of impairment, can present additional challenges for public transport systems aiming to support the dignity of people with disability through inclusion and accessibility (Kamruzzaman et al. 2016; Románach and Lobato 2005). Accessible and inclusive public transport enables people with disability to move freely and engage in employment (Jónasdóttir, Egilson, and Polgar 2021), education, socialisation, cultural activities, leisure, (Calle et al. 2022; Crudden, Cmar, and McDonnall 2017), receipt of services (Fortune et al. 2020) and improved access to medical care (Badji et al. 2021). The converse is also true. Lack of access to transport and constrained personal mobility can lead to social exclusion, social isolation, loneliness, poverty, poorer physical and mental health, and deprivation (Burholt, Windle, and Morgan 2017; Kamruzzaman et al. 2016). For people with disability, freedom of movement – and subsequently, perceptions of dignified mobility experiences – is inextricably linked with access to public transport that affords equal opportunity to dignified and self-determined participation in society.

Freedom of movement and dignified mobility experiences require accessible, inclusive, mobility-relevant information and communication practices across time and space (Bigby et al. 2019; van Holstein, Wiesel, and Legacy 2022). Barriers to freedom of movement include physical exclusion at the intersection between impairments and environments that lack accessible and/or inclusive services and facilities; geographical exclusion particularly in regional and remote areas; economic exclusion; and psychological exclusion including feelings of fear, indignity, and lack of acceptance (Calle et al. 2022; Church, Frost, and Sullivan 2000; Levine and Karner 2023). Inequality in the accessibility and availability of information, and difficulties related to the physical access and navigation of transport structures, also hinder dignified mobility experiences for people with disability (Almada and Renner 2015; Park and Chowdhury 2018, 2022; Velho 2019). Interpersonal, ableist discrimination can further impede dignified mobility experiences through impacts on safety and safety infringements. (Calder-Dawe, Witten, and Carroll 2020; Wayland et al. 2022; Velho 2019).

Perceptions of human dignity are characterised equally by the absence of discrimination, shame, and humiliation, as they are by positive experiences of acceptance and inclusion (Hojman and Miranda 2018). Thus, when

freedom of movement is diminished by physical, social, and sensory barriers that hinder equitable and independent participation, the dignity experience for people with disability is inevitably compromised. In contrast, successful engagement, acceptance, safety, responsiveness, and involvement in society by enabling freedom of movement through accessible and inclusive public transport services may represent a powerful bridging system capable of promoting dignified mobility experiences for people with disability. Facilitating equal access and opportunity for people with disability using public transport increases their capacity for self-determination and facilitates a greater sense of inclusion and participation in society (Kamruzzaman et al. 2016; Smith 2013).

Exploring and understanding the dignity experiences and perceptions of people with disability using public transport services will allow for greater insights into how dignity is conceptualised by people with disability, and the factors that contribute to dignified inclusion in society through freedom of movement. To decrease the prevalence of challenges associated with using public transport, and subsequently eliminate barriers to personal mobility for people with impairments, it is important to examine the interaction between public transport experiences and perceptions about dignity. Most studies exploring public transport systems have been conducted with participants without disability, the findings of which may not translate accurately for the disability community (Park and Chowdhury 2022). There is limited knowledge about the relative importance of access and access barriers across entire journeys, especially from the viewpoint of people with disability (Calle et al. 2022; Levine and Karner 2023; Park and Chowdhury 2022). By better understanding the link between public transport and dignity, we expect to improve understanding about the barriers and enablers to dignified mobility experiences. We also expect to increase understanding about factors associated with freedom of movement that influence and impact accessibility and inclusion of people with disability in interaction with public transport and society.

This research aimed to understand how dignity was experienced by people with disability, in interaction with trains and buses in an Australian city, and how dignity may impact freedom of movement by the ways people choose – or do not choose – to interact with trains and buses. Barriers and enablers to dignified mobility experiences were examined. Diverse and heterogeneous mobility experiences were sought to examine dignified mobility experiences specifically in interaction with buses and trains.

Research process

This research was approved by the Griffith University Human Research Ethics Committee (HREC) reference number 2020/939. Qualitative research design

comprising semi-structured in-depth interviews and framework analysis (Ritchie and Spencer 1994, 2002) underpinned this project. People with disability were asked specifically about their use of trains and buses in an Australian city and how their transport experiences impacted their personal mobility and dignity. The macro public transport system in this Australian city includes a variety of conveyance options, including ferries, buses, trains, light rail, taxis, ride-share, e-scooters, and so on. Trains and buses were the primary focus of this research because they carry larger numbers of people and were identified as conveyances where gaps in service experience can occur for people with disability. Participants were asked specifically about their experiences accessing, using, and riding on trains and buses, as well as about their general interaction and interfacing with the broader public transport system products and infrastructure (digital signage, journey planning apps, customer service etc.) Data were collected between January and March 2021, with careful consideration to ongoing public health restrictions related to COVID-19.

Sample

For the purpose of this research, disability was defined as ‘the interaction between persons with impairments and attitudinal and environmental barriers that hinders full and effective participation in society on an equal basis with others’ (CRPD (Committee on the Rights of Persons with Disabilities) 2006, 2). However, participants were asked to self-identify the language and terminology that best described them. People with disability were purposively recruited to this research *via* an introductory recruitment email and flier sent to people who had previously consented to being contacted for research opportunities by a public sector organisation in Australia. The public sector organisation also advertised the research in a regular newsletter sent to public transport users who had opted into receiving the newsletter. People with disability were eligible to be included in this research if they: (1) self-identified as having visible or invisible impairment; (2) had experience using public transport; (3) were aged 18 and over and (4) were able to consent to participating. Interested participants completed an online demographic survey, to support purposive sampling. Participants may or may not have been regular public transport users.

Seventy-five (75) people expressed interest in participating in the research and completed a demographic intake survey. Thirty (30) participants were purposively selected; however, one person did not respond to further follow-up, one became ill, and two decided not to participate after scheduling an interview. These participants were not replaced, because all other people who had expressed interest had already been informed about their exclusion from the study. Consequently, 26 people participated in interviews

ranging from 25 min to 1 h and 10 min, with the average interview time lasting 38 min.

The sample included 13 participants that identified as female, 12 as male, and one as non-binary. The median age bracket was 40–49 years old, with most participants aged 30 years and over (88%). Sixteen participants were regular transport users, six used public transport sometimes and four rarely used public transport. There was a near even split between participants of visible ($n=17$) and invisible impairments ($n=18$). The number of people with visible and invisible impairment was greater than the number of participants, because some participants identified as having both impairment types. Most participants identified as having a physical impairment ($n=21$), the next largest category being chronic pain or illness ($n=9$). Participants could select multiple impairments to best describe their individual circumstances.

Data analysis

Demographic data were analysed descriptively. Interview data were thematically analysed using Framework Analysis (Ritchie and Spencer 2002). Framework Analysis Method is grounded in the accounts and observations of participants in an inductive manner, but also supports deductive inclusion of aims, objectives, and a priori literature (Pope, Ziebland, and Mays 2000). Framework Analysis follows a five-step process that focuses on contextual, in-depth systematic analysis: (1) familiarisation; (2) thematic framework; (3) indexing; (4) charting; and (5) mapping and interpretation (Ritchie and Spencer 2002). During familiarisation, KC immersed themselves in the diverse data through reading specific interview transcripts and noting recurrent themes. KC and CE then established coding nodes, using a combination of indicative themes, a priori issues, and emergent issues raised by participants (Ritchie and Spencer 2002). The framework was provisionally applied to transcripts and refined as required. During step 3, indexing, KC applied the nodes to all interview transcripts using the NVIVO™ software package (QSR International Pty Ltd 2020). Data were specifically interrogated to ascertain structural barriers that impacted the relationship between dignity experience and public transport. Emerging themes were discussed with the wider research team at regular intervals. Charting, step 4, was conducted both collaboratively with the research team and specifically with CE. In charting, data were placed into headings and subheadings to build a thematic framework, with specific reference to original participant text (Ritchie and Spencer 2002). In the final step, KC presented a larger data map of all emergent categories, associations, and patterns to the research team for discussion, amendment and establishing final range of results (Ritchie and Spencer 2002).

Results

Dignity mobility experiences on buses and trains

Participants were asked to define dignity generally and in the context of their experiences and interactions with various public transport systems, notably buses and trains, although some participants spoke about taxis, light rail and ferries. Participants described dignity as 'maintain[ing] respect and self-worth' (P004), 'being treated like everyone else' (P010) and related to feelings of independence. Participants identified that dignified mobility experiences were not isolated or momentary experiences. Rather, entire travel journeys that were accessible, inclusive, equitable, promoted independence and enhanced self-worth contributed to dignified mobility experiences. Although some aspects of dignified mobility experiences related to transport infrastructure and conveyances, many more related to interpersonal interactions that were experienced in physical, digital and communication spaces across entire travel journeys. Positive interpersonal interactions included feeling respected, being acknowledged, being appropriately helped, belonging, and being treated like anyone else. Therefore, for people with disability, dignified mobility experiences are not confined to tangible infrastructures such as conveyances, stations and platforms, but are also influenced by intangible features that extend beyond buses and trains within transport systems to include societal factors such as culture, people, attitudes, values and feelings.

Acknowledgement

Through examining dignified mobility experiences, acknowledgement emerged as a critical enabling factor. Participants wanted to be acknowledged as a person first, with value, humanity, and the inherent right to interact with and use buses and trains in ways that supported their dignity. Participants shared that they wanted to 'just [be] treated as a person first' (P021) and be 'accepted and respected for who I am and what I am' (P019). Acknowledging, recognising and accepting participants' personhood enabled a dignified mobility experience, even when other aspects of public transport systems were not universally accessible and inclusive. There were three primary enablers by which participants felt acknowledged when using public transport systems that enabled dignified mobility experiences: (1) independent and easy access; (2) accessible, inclusive and personally relevant information; (3) respectful and empathic attitudes.

A note on individual factors

Individual factors, like socio-economic status, impairment type, and familiarity with public transport systems, influenced participants' perceptions

of dignity. There were some differences in how participants with invisible or multiple impairments perceived dignity because of the cumulative complexity of their individual circumstances. Participants with invisible impairments experienced more barriers related to access and attitudes, while participants with visible impairments noted members of the public staring or treating them in ways that made them feel vulnerable. While individual factors were integral to participants' interpretations of dignity, the primary focus of this research was to examine the ways in which dignified mobility experiences for people with disability could be enhanced or hindered by their interaction with public transport systems. Therefore, the results and discussion sections focus mostly on the way that public transport systems can uphold dignity for people with disability, rather than on the individual, personal and subjective influences on each interaction for each individual.

Acknowledging through enabling independent access

Dignified mobility experiences were first determined by the ease of independent access to all parts of transport journeys. Participants felt that their humanity was acknowledged when their needs were addressed, and they felt included and enabled to independently access both physical and virtual aspects of trains and buses. The ability to easily access the physical environment, including stations and stops, boarding and disembarking conveyances, and necessary facilities increased participant's perception of dignified mobility experiences. Access to the virtual environment included websites, apps, digital signage, and announcements.

Participants closely correlated the ability to access various components of public transport systems with independence and dignity. Participants said that 'dignity is independence... by allowing us to be more independent, you're giving us dignity' (P018). Independence was a critical part of dignified mobility experiences, mentioned by 15 participants (58%), as it contributed to increased feelings of safety, security, and freedom. For example, 'I just think that people with disabilities... need to feel like they are included and need to feel like they're really involved in decisions that might affect them' (P014).

Independence was predicated on universally designed, inclusive and accessible infrastructure across entire interactions with public transport systems. Independently accessible parking, seating, shelter, and toilet facilities enabled dignified mobility experiences and thus freedom of movement. Feelings of independence enhanced choice, spontaneity, and dignity. One wheelchair user said that independent use of conveyances made them feel 'human' and as though they could 'do what I want' (P013). Conversely, lack of access contributed to some participants' decreased confidence or willingness to

navigate a travel journey alone, thereby decreasing freedom of movement, inclusion in society and independence.

Minding the gap

Dignified mobility experiences were influenced by the difference between participants' expectations for how they might interact with public transport systems, and the reality of that interaction, especially related to accessible components of a system. Participants discussed the need to mind the gap – theoretically and physically – noting that there were more barriers to people with disability than those without. Some gaps in experience or service were known about prior to undertaking a journey while other gaps emerged during a travel journey, through disruptions and other unexpected interactions, like a station master not being ready with a ramp for disembarking.

Minding the physical gap related mostly to boarding and disembarking trains and buses which was mentioned by 20 participants (76%). Participants experienced varying levels of difficulty with boarding and disembarking:

Getting in and out...doesn't help with my dignity, because a lot of time it's [the bus step-off point is] very high and I've got to step right down [onto the curb]...I've got to always hold on to the sides [of the steps] to get out and that also makes me feel like people are staring at me; I'm taking too long because everyone's always in a rush behind me (P007).

Four participants (15%) who identified as wheelchair users shared experiences in which they were left on trains because they could not independently disembark. Rather they required staff to manually place a ramp for disembarking. All four participants commented that they felt entirely undignified and vulnerable during those experiences.

Minding the theoretical gap related to differential access for people with disability and those without, in precincts and on conveyances and when using ticketing products. Precinct environments and conveyances had varying levels of accessible and inclusive design, facilitating independent access. Some environments described and utilised by participants were only inaccessible to people with certain types of impairments, some had limited access, and some were designed for universal access. For wheelchair users, varying access levels at precincts increased vulnerability to undignified mobility experiences, particularly when 'a platform is accessible but it's only one way. So, you can get off on one side but the other side's not accessible' (P013).

Comfort and safety also impacted on dignified mobility experiences for some participants, particularly through access to seating and shelter. Seemingly small precinct design issues, like toilet doors with handles instead of push buttons or lack of an accessible boarding place, increased participants' feelings of indignity and decreased overall comfort with travelling.

However, participants felt mostly comfortable and safe when accessing precincts and using conveyances.

Ticketing and ease of accessing both ticketing products and ticketing validators were mentioned by 18 participants (69%) with varying impairments. Independently interacting with ticketing validators proved difficult for many participants. A range of impairment-specific factors and positional factors of validators at precincts and on conveyances contributed to these difficulties. Participants with cognitive impairment experienced increased vulnerability to undignified interactions when they could not remember to tap off on a validator, resulting in higher fare charges.

The good news

Despite experiencing access barriers that resulted in increased vulnerability to undignified mobility experiences, participants noted that, overall, public transport systems are greatly improved in the past 10–15 years. Participants appreciated that ‘now every station has a board [mobile ramp] to get onto the train and they’re easily accessible by the staff’ (P010). Improved access showed participants that they were acknowledged as valuable members of society and felt that their right to use public transport was maintained. Overall, participants felt that access was improving and that the most significant barriers to dignity came not from access but from ‘the treatment you receive’ (P015), which impacted the way they felt acknowledged when using public transport and will be discussed in detail below.

In summary, dignified mobility experiences were first determined by the ease of independent access to all parts of transport journeys that included public transport systems. Participants felt acknowledged as valuable members of society and that their human rights were upheld when they could easily and independently access the transport they desired. Participants identified design barriers of infrastructure and environments, ticketing and associated products, and boarding and disembarking as particularly burdensome to their dignity experience.

Accessible and inclusive information as a form of acknowledgement

When participants had previously experienced barriers to independent access and interaction with buses and trains or anticipated that they might experience barriers on their next journey, they endeavoured to pre-empt and pre-plan how they might experience their travel journey. Accessible, inclusive, transparent, accurate, and consistent information enabled participants’ dignified mobility experiences. Participants felt acknowledged and valued when they were able to obtain information, they needed to accurately make sense about their travel journey. Dignity at the intersection with information was

closely related to independence, in so far as participants could access, assess, and use information to support independent navigation and interaction with public transport.

Dignity is me having the skills to do it myself, but there also being appropriate messages. So, say for the visually impaired, it's voice or audio. For deaf, having good signage. And same for those with intellectual impairment, having very plain language signage and not relying on icons (P018).

Twenty-one participants (81%) discussed seeing, reading, hearing, and understanding information. Information about buses and trains was available online, *via* signage in precincts, through announcements and interaction with staff and other travellers. Participants recognised that most information was accessible and inclusive, '[trains] do have visual announcements to tell you where the next stop is or what the next stop is' (P003). Participants discussed the need for visible and visual, auditory and audible, easy to understand and real-time information to support dignified mobility experiences.

Visible and visual information

Accessible and inclusive visual information that improved dignified mobility experiences referred to placement, size, and simplicity of information. Signage was frequently mentioned as both a hindrance, when not visible or accessibly presented, but also as an enabler for improving participants' ability to navigate trains and buses more easily and with increased dignity. The visibility of route designation is essential for participants to independently identify the correct bus or train for them. In the case of buses, early detection of bus route numbers is necessary, because buses do not necessarily stop in a pre-determined position at large and busy bus stops, meaning that participants may need to quickly move to a different position to board. One participant with vision impairment mentioned a negative change from previously accessible large, high contrast print to slim-line signs as a frustrating barrier:

Back in the day, there used to be these great big white signs [on buses] that would say [destination] or whatever. So that way even if the sound [audible information] wasn't working you could at least see the first big letter, even if you couldn't read the word properly... now they've put these itty bitty little slim line 'look at me, I'm sexy' [signs, which are difficult to read] (P015).

For participants who were Deaf, visible signage and readable information was critical to a dignified mobility experience, particularly when circumstances changed during travel journeys:

I am not sure what was happening. I did not know what was going on...I look at people's body language, so I can pick up what is going on when a train has

stopped for some reason. If I can see people are panicking or making phone calls to everybody, then I might know what is going on. I then realise that the train has stopped for some reason, but I don't know why there is a delay. That is what happens for me (P003).

Participants who were Deaf or hard of hearing commented that:

[Trains] do have visual announcements to tell you where the next stop is or what the next stop is, but that is the only type of visual announcements they have... with hearing aids on I can hear that there is an announcement, but I don't know the words that they are saying (P003).

Lack of information that is visible and visual made it difficult or impossible for participants who were Deaf to access the information required to sense-make about the next steps on their journey, increased participant anxiety and vulnerability to undignified mobility experiences.

Audible and auditory information

Participants discussed the need for accessible and inclusive information in audible and auditory formats to ensure dignified mobility experiences, especially for participants with low vision. One participant with low vision mentioned the impact that lack of accessible audio information had on their ability to navigate travel journeys and said, 'I've hailed trucks before. I've hailed wrong buses. I've gotten on wrong buses. Yeah, if the sound system at the train station isn't working, then I don't know what train's coming' (P015). Lack of auditory communication placed participants who were blind or had low vision in vulnerable situations and increased their anxiety.

Easy to understand information

The content of information and how it was presented was discussed by participants as both a hindrance and an enabler to dignified mobility experiences. Participants with mental health and cognitive impairments mentioned the difficulty they experienced when reading information that was not presented in plain language. Journey and route maps were particularly difficult to engage with and understand, 'all the different options and routes and colours and everything. I understand that you've tried to simplify it as much as you can, but it's just at least for me, it's still a lot to get around [comprehend]'. (P011). Another participant mentioned that the use of icons, typically used to improve the clarity and brevity of communication and information, was difficult for them to understand:

'Some icons I find really hard. Because if there's a lot of detail in the icon, I can't work out what it is. You can get, some icons, obviously male and female at toilets, they're good... If you've got too many icons, where they might be slightly similar, but convey different meanings, it can be confusing' (P018).

Information that was accessibly presented in multiple formats is a key underpinning principle of Universal Design. It increased participants' ability to make choices about navigating public transport journeys and choose how they wanted to interact with the system to enhance their dignified mobility experience.

Accurate, real-time information

Accessible and inclusive information went beyond hearing, seeing, reading, or understanding information and included real-time, accurate information. Participants with impairments that impacted mobility, particularly wheelchair users, required accurate information to determine whether a mobility experience would be accessible and dignified, 'I don't know if the train stations are accessible. I kind of, sort of, know which ones are' (P016) and this knowledge was developed through experiential trial and error, creating high stress situations, which resulted in undignified experiences.

Real-time information was essential for participants, especially while travelling when journeys were disrupted. Participants noted that planned travel disruptions were often well communicated, early and advanced, in ways that were accessible. However, unplanned and emergent journey disruptions, even just the threat of disruption, impacted most on dignified mobility experiences for participants:

All the anxiety just keeps going until we actually arrive [at planned destination] because it's not like I can get to [station] and then walk up the stairs because that's just not going to happen. So, it's the added hindrance of always being on edge and always being on nerve that something's going to change (P027).

When accessible communication occurred in real-time, participants experienced decreasing levels of anxiety and increasing levels of dignity.

Sense-making and pre-planning

Nineteen participants (73%) undertook comprehensive pre-planning and sense-making about their travel journey, to help manage anxiety, retain control over their mobility experience, and navigate transport interactions effectively and with dignity. Pre-planning and sense-making required access to timely, accurate, consistent, and reliable information, which participants sometimes used to create personalised navigation journeys. For example, 'You've just got to plan where you want to go... like it's that whole holistic journey approach, so from when you leave home to where you've got to get to your destination' (P013). While pre-planning and accessing information prior to travel journeys and during travel to support dignified mobility experiences was beneficial, some participants described frustration and feelings of indignity about the need to do extra sense-making over and above

what people without disability may need to do, to experience independent access.

Participants consistently recognised the necessity for journey planning. When participants were confident and experienced at using transport routes, they were less anxious about their journey and more likely to navigate the system easily while maintaining dignity. However, all participants wanted to familiarise themselves with where to go and what to do during journeys. Even regular routes needed to be learned. Participants engaged in sense-making activities by acquiring knowledge about intricate and personally-relevant travel details including where to find lifts at train stations, and where they needed to be to most conveniently board the correct bus or train. Participants gathered information and details about where amenities were located and whether amenities were accessible to them, parking availability and accessibility, and whether they felt comfortable and safe in travel spaces. Journey planning also involved timetabling and route planning.

In summary, information was a critical enabler of dignified mobility experiences for participants and was a way that public transport systems could acknowledge participants as customers with worth and value. Access to information enabled participants to sense-make and plan their travel journeys and enabled dignified mobility experiences when elements of journeys changed in real-time. Visible and visual, audible and auditory information enabled participants with diverse impairments to access information in ways that best suited them. Easy to understand information that was communicated in real-time decreased anxiety, especially during travel disruption, and increased dignified mobility experiences.

The contribution of personal interactions to dignified mobility experiences – acknowledgement and consideration

Dignified mobility experiences for participants were enabled by interpersonal interaction with staff and other customers. Participants felt acknowledged and dignified when staff engaged respectfully and empathetically. Staff attitudes and awareness towards impairment, staffs' ability to engage with empathy, and staffs' role in help-seeking were critically important for enabling dignity for all participants. Participants shared many positive experiences where frontline staff acknowledged and treated them like any other person. Staff could improve dignified mobility experiences by ensuring 'my wants and needs are respected and that I'm not treated like some inspiration or burden' (P016). Participants generally felt that public transport staff '...come across warm and friendly and I feel supported that if there was any issue that I could actually get help if I needed it' (P006). Conversely, participants reported that staff attitudes could inhibit dignified mobility experiences, 'They [staff] just sort of look at you and go "Oh my god, it's someone with

a wheelchair, what do I do?" Like, mate, you can talk to me, you're not going to catch my wheelchair disease' (P013). Undignified interactions with staff led to participants feeling dependent, experiencing increased anxiety and emotional distress; increased social isolation; and decreased feelings of dignity. When staff attitudes were impersonal or dismissive, participants felt susceptible to undignified mobility experiences, the impact of which was best highlighted by participants who were wheelchair users and often more dependent on staff assistance to navigate accessing trains and buses:

They [hospital staff] said to me when I was in hospital, 'You'll get your dignity when you leave.' It's like, 'No, I don't', because everyone knows my life but you've got to explain things over and over again. When you plan your days, you are jumping on a train, you have to let the staff know where you're going ahead of time so that they can plan ahead, have a guard there to get you on and off the train (P013).

Help-seeking and relying on staff for assistance was both an enabler and inhibitor to dignified mobility experiences. Most participants acknowledged that staff were empathetic, kind, and had good intentions, 'the staff are always amazing... they're always there and you can see that they're looking out for people and they're checking everyone' (P007). Participants wanted staff to acknowledge that they were just like other public transport users without impairment. During help-seeking, participants reported that staff often pre-emptively offered assistance but were respectful when the assistance was declined, which promoted dignity:

Dignity means to me then, having the respect to say to me, how can they help you. Or actually even first off, 'Do you need help?' and acknowledging that I might not need help. I will ask for help if I need it, but I may not need help (P018).

Participants also reported that staff attitudes during help-seeking could inhibit dignified mobility experiences. Participants wanted staff to acknowledge and accept that their impairment may impact their ability to navigate the system but did not want to be asked for additional proof or disclosure about individual impairments. Participants with hidden impairments described continuously disclosing their impairment and related information for staff to adequately help them and acknowledge their individual preferences. The burden of providing proof or details left some participants feeling very undignified, 'I shouldn't have to go up [to the bus driver] and say, "I've got crappy [bad] knees. Here's all the scars to prove it. Please let me off the front." That takes away my dignity because I feel like I'm begging to be treated with respect like everybody else on that bus' (P026). The dignified mobility experience related to staff interactions, or lack thereof, was also impacted by COVID-19, where changes in policy and procedures and the subsequent inconsistent application of those procedures required participants to ask for more assistance.

The attitude and perceptions that staff held about participants and the way that staff interacted and engaged with participants influenced whether dignity or indignity was experienced. When participants felt compelled to unnecessarily prove and justify their needs, they felt undignified. Conversely, when staff interacted with participants in ways that were consistent and empathetic, participants felt respected and dignified.

Discussion

Dignified transport systems are universally designed without gaps, but that's not reality

A transport system that includes buses and trains and that ensures dignified mobility experiences for people with disability looks like one that is universally designed without gaps. However, that is not the current reality of most global transport systems, as demonstrated in this research and other recent studies (Levine and Karner 2023; Mathews, Marshall, and Wilkinson 2022; Power 2016). The legacy of the medical model of disability in transport planning and economic prioritisation makes it difficult for Universal Design to be achieved. Most transport planning occurs without engaging marginalised populations, particularly people with disability, resulting in ongoing challenges and gaps for customers (Levine and Karner 2023; Park and Chowdhury 2022; van Holstein, Wiesel, and Legacy 2022). Participants in this research commented on the need for increased engagement of people with disability in planning, design, and development of products, conveyances, infrastructure, and services that they will use in interaction with public transport systems, particularly buses and trains.

Transport systems have developed and endured over time, making existing infrastructure more immutable than the societal values that existed when they were built (Spray et al. 2022). Although newly-constructed transport precincts and conveyances can and should more readily adopt Universal Design, retrofitting older, established parts of transport networks are costly and likely to be constrained by existing infrastructure – these issues will not be overcome easily (Park and Chowdhury 2022). Even when Universal Design principles are applied, they are often done so inconsistently, creating additional gaps in services and expectations, making it increasingly more difficult for people with disability (Calle et al. 2022; Grise et al. 2018; Starzyńska et al. 2015). Accessible and universal design elements, like buses with low-floor access, were present in the conveyances and infrastructure accessed by participants in this research; however, broader system constraints and limitations continue to present and reinforce barriers for people with impairments (Power 2016).

A critical impediment for the proliferation of Universal Design in transport systems is not just transport planning and immutable infrastructure, but a lack of legislative and policy imperative (Levine and Karner 2023; van Holstein, Wiesel, and Legacy 2022). Signatories of the UNCRPD agreed voluntarily to the charter; however, consequences for not meeting human rights obligations are few. In the United States, the Americans with Disabilities Act (1990) relies on 'voluntary compliances and allows agencies to opt out of providing accommodations if they would be financially burdensome' (Levine and Karner 2023). Similarly, in Australia, most states are well behind in achieving compliance with the *Disability Standards for Accessible Public Transport* (AHRC (Australian Human Rights Commission)) 2002), which required 100% of bus stops be compliant by 31 December 2022.

Compliance based initiatives, like legislation in the United States and Australia, further highlight ableist tendencies, where broad based standards are assessed as having been met, but the individual intricacies of meeting those standards for people with impairments are not universally met because their diverse needs are not well understood nor resourced (Levine and Karner 2023; Mathews, Marshall, and Wilkinson 2022; van Holstein, Wiesel, and Legacy 2022). Mathews, Marshall, and Wilkinson (2022) highlighted that universally accessible standards were helpful, when met, and solved common issues for many people with impairment, but not all. The people who were not well served by those standards experienced additional exclusions and stigma, as they were not able to engage with or utilise their respective transport services, which 'contravenes the citizenship right to dignity, empowerment, and autonomy' (Mathews, Marshall, and Wilkinson 2022). These types of experiences were highlighted throughout this research, whereby participants with diverse impairments found similar products, services, and infrastructure elements inaccessible but for different reasons. Policy changes and embedded mandates for including people with diverse impairments in planning and development is urgently needed to redress the structural barriers that exclude people with disability (Calder-Dawe, Witten, and Carroll 2020; Levine and Karner 2023; Stafford and Baldwin 2018).

Gaps in design increase across journey chains and at intersections with multiple systems

The reality of theoretical and physical gaps was articulated by participants in this research, who identified numerous pain points and challenges in achieving a dignified mobility experience when using trains and buses. Participants commented that transitions between intra-systemic elements (e.g. train station to train platform) and between inter-systemic elements (e.g. train to bus; public transport to private transport) increased the

likelihood of gaps and undignified experiences. Transport systems are not singular systems used in isolation, but rather a network of micro, meso, and macro systems that intersect, overlap, and integrate in ways that can promote seamless use or increased gaps and challenges (Vertesi 2014).

Journeys through transport systems are complex, comprising different modes, routes and providers across an entire transport network (van Holstein, Wiesel, and Legacy 2022). Participants used both wayfinding and wayfaring to navigate theoretical and physical gaps, that were both known and unknown prior to departing. Wayfinding encompasses planning and deliberate prior sense-making to establish an expectation for how a journey might unfold or identifying potential known gaps (Prescott et al. 2020). Wayfaring is much more complex and requires people with disability to navigate their expectation about how a journey might unfold versus the actual experience, or navigating emerging gaps (Prescott et al. 2020). In the main, this research explored the use of trains and buses; however, most participants discussed how individual elements of these components of journeys linked together to create a holistic transport journey, or a journey chain (Park and Chowdhury 2022). For a journey to be dignified, the entire journey chain should be accessible (Henly and Brucker 2019; Park and Chowdhury 2022). This research highlighted that aspects of train and bus transport were often inaccessible. Thus, the reality is that fragmentation across different elements of transport systems result in less accessibility, inclusion and dignity for people with disability (van Holstein, Wiesel, and Legacy 2022). The more complex a customer's journey chain is, the more prevalent and problematic the gaps become, the more they must depend on wayfaring through gaps, which increases anxiety.

Creating empathic systems to enhance dignity and respond to gaps

While planners, policymakers, and transport leaders work toward Universal Design and a seamless transport system, participants in this research articulated approaches that acknowledge and highlight actual and potential systemic gaps in a journey chain while enhancing dignity for people with disability. Dignity is about being acknowledged and having human rights met in ways that are personal, timely, and responsive to preferences. Enhancing and promoting dignified mobility experiences can be achieved by creating empathic systems, even in the presence of gaps. Empathic systems are ones where information and communication are delivered in ways that are responsive to people's needs and preferences, while the attitudes of frontline staff can be used to enhance dignity and deliver empathic responses, especially in the face of emerging gaps. Information and empathic attitudes can serve as bridges that circumnavigate gaps while still supporting the dignity experience of people with disability.

The importance of information

Information is the first critical element to acknowledging and enhancing dignity for people with impairments in public transport systems. Information needs to be known, available, and accessibly communicated. Going beyond the journey planner, information should be timely, personally relevant, and informed by lived experience data (Levine and Karner 2023). Access to information enables people with impairments to make decisions about the ways in which they choose to interact or not interact with public transport systems. Information can be used to sense-make and navigate public transport systems as independently as possible, which increases feelings of safety, support, connection, and autonomy for people with disability (Darcy, Maxwell, and Green 2016; van Holstein, Wiesel, and Legacy 2022).

Participants used information to support both wayfinding and wayfaring across entire journey chains. Participants conducted extensive pre-planning during, which required access to reliable and accurate information, communicated in multi-modal ways that were accessible and inclusive. When information could be successfully gathered during wayfinding, participants felt more confident in navigating public transport systems, understanding where gaps were likely to appear and potential solutions for circumnavigating those gaps.

The importance of accessibly communicating information was highlighted during wayfaring, as participants were required to navigate through gaps that emerged during travel in service and experience. Gaps were often a consequence of lack of information communicated prior to commencing a journey. Wayfaring increased participants anxiety and vulnerability to undignified experiences.

People with disability need access to complete, transparent, visible, audible, and easy to understand information to enhance dignity and feel confident when navigating a public transport system. Information should be easy to find prior to travel and during wayfaring. It should be consistent and accurate and provided as close to real-time as possible. Most importantly, information and the way it is communicated should be informed by lived experience of visible and invisible impairment, rather than on non-disabled and medical model assumptions about how people (usually those with visible impairments) interact with transport systems (Levine and Karner 2023). Digital information and communication solutions are already used by many public transport systems internationally. However, over-reliance on smart phone accessible technology, typically developed without consultation from people with diverse impairments, can further increase the gaps for marginalised populations who may not have or use smart phones. Achieving accessible digital information and communication solutions requires inclusion of people with disability during solution design phases.

Empathic staff: a key role

Developing environments that support staff to be engaged and empathic is essential to acknowledgement and dignity and the second critical element for overcoming gaps in public transport systems. Empathic staff are critically important for changing the overt and covert barriers that people with impairments encounter when interacting with public transport systems. They can support wayfaring with additional information while assisting people with impairments, in the ways that are dignified, with navigating around gaps. However, overcoming the proliferation of ableism requires staff training and development to enable staff to engage confidently, in dignified ways with customers with impairments.

Staff, as echoed in both this research and contemporaneous literature, play a critical role in the covert, mundane ableism and associated barriers that are experienced by people with impairments (Calle et al. 2022). Frontline staff, particularly those in customer service in public transport systems, can be viewed as street-level bureaucrats, who must deal with policy and practice gaps while negotiating the variable demands of public transport systems (intra-systemic and inter-systemic) (Bitner, Gillett, and Foster 2023; Hummell et al. 2023; Lipsky 1980). Staff may not have the capability, opportunity, or motivation to engage with dignity. High levels of empathy are based on 'ecosystems of respect' that permeate the entire fabric of service and social systems (Kendall and Barnett 2015). The goal of this systemic empathy is to develop general concern and positive regard for those that have been treated the worst over time and to do so in a dignified manner (Zimmer 2003). Empathy has been found to be associated with improved staff well-being (Melnick and Powsner 2016) and higher levels of service user enablement (Derksen, Bensing, and Lagro-Janssen 2013; Hojat et al. 2013; Jeffrey 2016; Neumann et al. 2009). Co-designed training that extends beyond box-ticking diversity and inclusion training is essential to ensure that staff can engage empathically to maintain the dignity of all customers.

Achieving systemic empathy requires further research that explores systemic influences on empathic behaviour. There must be an emphasis on: (1) identifying and measuring systemic factors that contribute to empathy and dignity; (2) implementing system wide empathy by learning rather than judging; and (3) recognising and understanding how to operationalise system wide empathy effectively and efficiently across diverse populations (Sokol-Hessner et al. 2018). Engaging more fully with human experience (Kendall and Barnett 2015) can support systemic empathy and as a result, the acknowledgement and dignified mobility experience of people with disability.

Limitations of the investigation

The study collected data about participant's experience of buses and trains as well as broader public transport systems in an urban/suburban context in a city in Australia. While this city does have ferries and light rail in

suburban locations, buses and trains are the predominant mode of public transport and were selected for feasibility and frequency of use, as well as identification of gaps in experience when using these conveyances. While a narrow definition of public transport was employed, participants did speak about their interactions with intra-systemic and inter-systemic elements of public transport systems in the city more broadly. Additional research should be conducted to examine other factors related to taxis, light rail, ferries, air travel and other potential modes of public transport. Finally, while the research team considered variables that can impact on travel, including regular vs. irregular use of transport and regular vs. irregular routes, additional factors like time of day and reason for travel should be examined.

Conclusion

Ensuring the dignified mobility experiences of people with disability when interacting with public transport systems is an increasingly important priority for Australian policymakers and service providers. Embedding practices that support people with impairments to access and feel included in transport systems ensures that Australia will meet their human rights obligations under the CRPD. People with impairments identified that dignified mobility experiences are closely related with feelings of acknowledgement and belonging. When public transport systems are accessible and inclusive to people with impairments, they felt their human right to access those systems were acknowledged, which resulted in increased dignified mobility experiences. It also took the 'disability' out of having an impairment as barriers were minimised. To ensure people with impairment feel acknowledged, barriers to use and access must be eliminated through the use of Universal Design in new construction and retrofitting older infrastructure when possible. When retrofitting is not economically feasible or infrastructurally possible, public transport systems should ensure that information is communicated accessibly and inclusively, so that people with impairments can sense-make about how they should best navigate the system, to support their own dignified mobility experiences. Finally, creating conditions for empathic systems will support staff to have the capability and opportunity to support the dignity of people with impairment across journey chains.

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Data availability statement

Access to de-identified data sets may be available upon request.

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